

Digital Image Processing

Mathematics of digital image processing

- The mathematics of digital image processing (DIP) involves representing images as arrays of pixels and applying mathematical operations like linear algebra, calculus, statistical methods, and Fourier transforms. These operations are performed on these pixel arrays for tasks such as enhancing, restoring, and analyzing images, which can include arithmetic and logical operations, neighborhood operations, and frequency domain transformations.

Image representation and basic operations

- ▶ Pixel representation: A digital image is treated as a 2D array of pixels, where each pixel has a value representing its intensity. For a grayscale image, this is often a single value, e.g., $I(r,c)$ for the pixel at row (r) and column (c) . For a color image, a pixel is an array of values, such as a triplet for RGB.
- ▶ Arithmetic operations: These are performed pixel-by-pixel and include addition, subtraction, multiplication, and division. For example, adding two images (F) and (G) results in a sum image (S) where $S(x,y)=F(x,y)+G(x,y)$. This can be used to reduce noise, as in averaging multiple noisy images.
- ▶ Set and logical operations: Operations like union, intersection, and complement are applied to images.

Advanced mathematical concepts

- ▶ **Linear and nonlinear operators:** A linear operator, (T) , satisfies $(T(a \cdot f + b \cdot g)) = a \cdot T(f) + b \cdot T(g)$, while a nonlinear operator does not. Image processing tasks can be viewed as applying an operator (T) to an ideal image (f) and adding noise, with the goal to find (f) from the noisy observation $(g = T(f) + n)$.
- ▶ **Spatial and frequency domain transforms:**
- ▶ **Spatial domain:** Operations are applied directly to pixels, such as intensity transformations and neighborhood operations.
- ▶ **Frequency domain:** Transforms like the Fourier transform are used to analyze an image in terms of its frequency components, allowing for frequency-based filtering and analysis.

- ▶ **Linear algebra:** Images can be represented as vectors and processed using matrix operations, which are particularly useful for transformations and feature extraction.
- ▶ **Statistical methods:** Concepts like conditional probability can be used to analyze relationships between pixels or image features

Applications in DIP

- ▶ **Image enhancement:** Techniques to improve visual quality, such as sharpening and contrast adjustment.
- ▶ **Image restoration:** Recovering a degraded image, often framed as an inverse problem where the goal is to estimate the original image.
- ▶ **Image segmentation:** Dividing an image into meaningful regions or object

Arithmetics & Logical Operations

- ▶ Addition
- ▶ Subtraction
- ▶ Multiplication
- ▶ Division



- And
- Or
- Not

Addition Operation

0	0	0	0
0	100	100	0
0	100	100	0
0	0	0	0

100	0	0	0
0	100	0	0
0	0	100	0
0	0	0	100

100	0	0	0
0	200	100	0
0	100	200	0
0	0	0	100

Subtraction Operation

0	0	0	0
0	100	100	0
0	100	100	0
0	0	0	0

100	0	0	0
0	100	0	0
0	0	100	0
0	0	0	100

0	0	0	0
0	0	100	0
0	100	0	0
0	0	0	0

Multiplication Operation

0	0	0	0
0	100	100	0
0	100	100	0
0	0	0	0

100	0	0	0
0	100	0	0
0	0	100	0
0	0	0	100

0	0	0	0
0	255	0	0
0	0	255	0
0	0	0	0



Division Operation

0	0	0	0
0	100	100	0
0	100	100	0
0	0	0	0

100	0	0	0
0	100	0	0
0	0	100	0
0	0	0	100

0	0	0	0
0	1	0	0
0	0	255	0
0	0	0	0

Logical Operation

▶ **AND**

▶ **OR**

▶ **NOT**

OR OPERATION

► Truth Table

0	0	0
0	1	1
1	0	1
1	1	1

OR Operation

0	0	0	1
0	0	0	1
0	0	0	1
1	1	1	1

1	1	1	1
1	0	0	0
1	0	0	0
1	0	0	0

1	1	1	1
1	0	0	1
1	0	0	1
1	1	1	1

AND Operation

► Truth Table

0	0	0
0	1	0
1	0	0
1	1	1

AND Operation

0	0	0	0
0	0	0	1
0	0	0	1
0	1	1	1

1	1	1	1
1	0	0	0
1	0	0	0
1	0	0	0

0	0	0	0
0	0	0	0
0	0	0	0
0	0	0	0

NOT Operation

► Truth Table

0	1
1	0

NOT Operation

0	0	0	1	1	1	1	0
0	0	0	1	1	1	1	0
0	0	0	1	1	1	1	0
1	1	1	1	0	0	0	0